

BattingEdge V9.5: AI Model Performance Report

Algorithm: Transformer Encoder (Self-Attention Mechanism) | Architecture: Multi-Head Attention

Date: January 13, 2026

1. Executive Summary

The **BattingEdge V9.5** Transformer model applied state-of-the-art "Self-Attention" mechanisms to the problem of cricket shot classification. Achieving an **Overall Test Accuracy of 81.75%**, the model demonstrates solid competency but falls short of the benchmarks set by the BiLSTM (91.8%) and XGBoost (87.3%) models.

This result validates a known principle in Deep Learning: while Transformers are powerful, they are "data-hungry" and often require significantly larger datasets to outperform inductive bias-heavy models like LSTMs on temporal sequences.

2. Methodology & Architecture

2.1 Model Design

- **Core:** Multi-Head Self-Attention (4 Heads). Unlike LSTMs which process frames sequentially ($t_1 \rightarrow t_2 \rightarrow t_3$), this model looks at the entire 50-frame sequence simultaneously to find global relationships.
- **Positional Encoding:** Added to the input to ensure the model understands the temporal order of the cricket shot (Stance vs. Follow-through).
- **Regularization:** Heavy dropout (0.3) and Layer Normalization were used to stabilize gradients during training.

2.2 Dataset Split

- **Training Set:** 3,007 samples
 - **Validation Set:** 388 samples
 - **Test Set:** 378 samples (Unseen data)
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3. Performance Metrics

3.1 Global Metrics

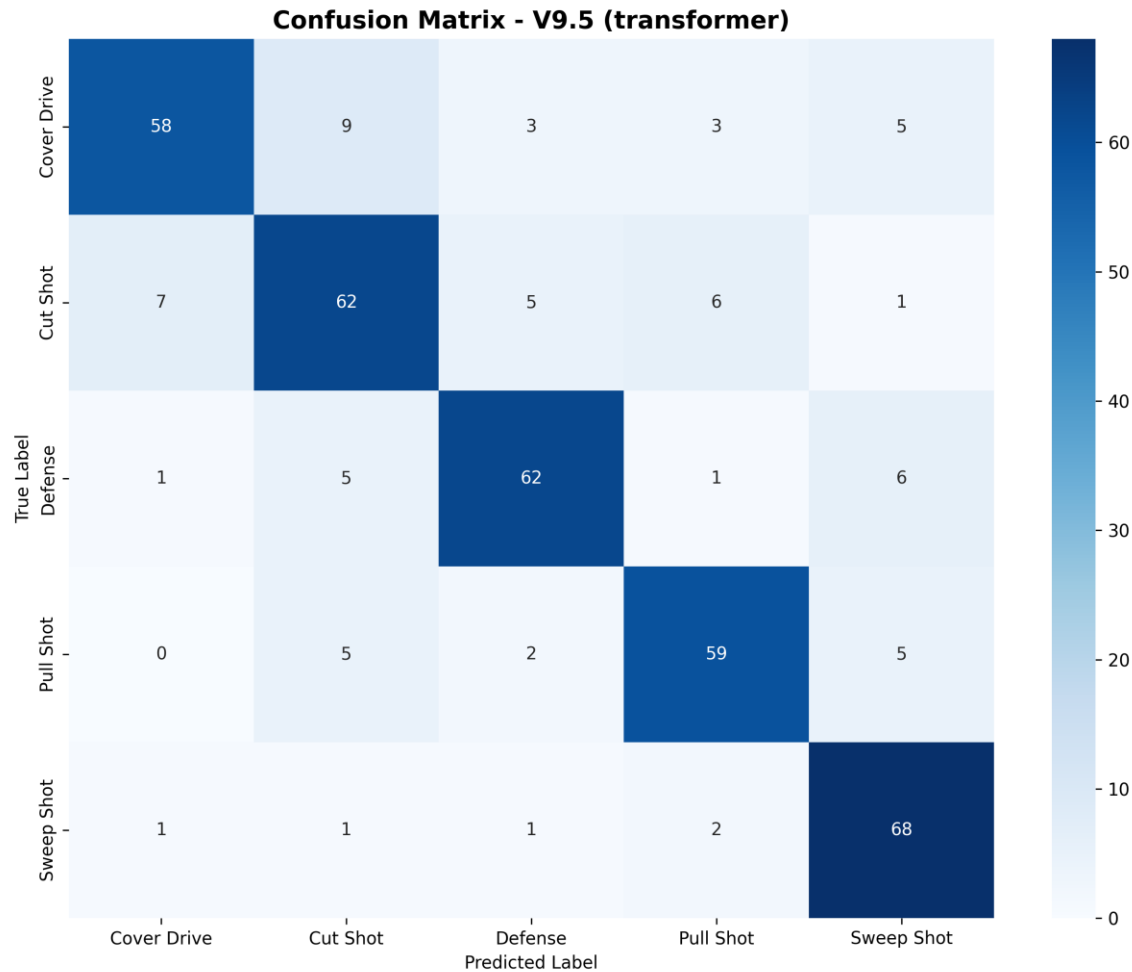
Metric	Score	Interpretation
Accuracy	81.75%	Competent performance, suitable for general use but lacks "Elite" precision.
Macro F1-Score	0.82	Balanced performance, though noticeably lower recall on Cut Shots and Drives compared to other architectures.
Convergence	Epoch 33	The model struggled to lower loss further, triggering Early Stopping.

3.2 Per-Class Analysis

The Transformer excelled at "distinct" poses (Sweep) but struggled with "similar" dynamic motions (Cut vs. Drive).

Shot Type	Precision	Recall	Accuracy	Assessment
Sweep Shot	0.80	0.93	93.2%	Elite. The distinct kneeling pose is easily captured by attention mechanisms.
Pull Shot	0.83	0.83	83.1%	Strong. Good recognition of the cross-bat motion.
Defense	0.85	0.83	82.7%	Solid. Reliable, though some confusion with cuts exists.
Cut Shot	0.76	0.77	76.5%	Average. Significant drop in performance compared to BiLSTM (87%).
Cover Drive	0.87	0.74	74.4%	Weak. High confusion rate; the model struggles to separate this from Cut shots.

4. Error Analysis (Confusion Matrix)



The Confusion Matrix reveals a specific weakness in the Transformer's ability to distinguish subtle wrist variations:

4.1 Primary Confusions

1. Cover Drive -> Cut Shot (16 total cases):

- *Analysis:* This is the most significant failure mode. The Self-Attention mechanism appears to focus on the "general width" of the shot. Since both shots can be played away from the body, the Transformer confuses the *Square Drive* with the *Cut Shot* more frequently than the LSTM.

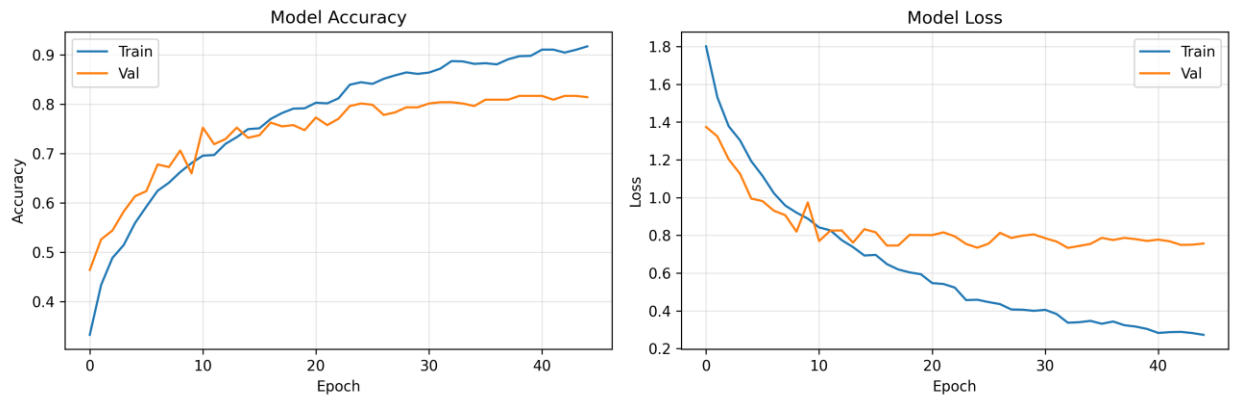
2. Cut Shot -> Pull Shot (6 cases):

- *Analysis:* Both are back-foot shots. The model correctly identifies "Back Foot" but fails to distinguish the bat path (Horizontal vs. Vertical) as clearly as the Angle-based RF/XGBoost models.

3. Defense -> Sweep Shot (11 mixed cases):

- *Analysis:* Likely caused by the positional encoding not strictly enforcing the "order" of movement as rigidly as an LSTM, causing confusion between a forward defensive lunge and a sweep entry.

5. Conclusion & Recommendations



Verdict

The Transformer is promising but not production-ready for this dataset size.

While 81.75% is a respectable score, it is significantly outperformed by the BiLSTM (91.8%) and Random Forest (89.4%).

Why did it underperform?

Transformers lack the "Inductive Bias" of LSTMs. An LSTM assumes that *frame 2* follows *frame 1*. A Transformer has to *learn* this relationship from scratch. With a dataset of ~3,000 videos, the Transformer hasn't seen enough examples to learn these temporal rules as effectively as the LSTM.

Report generated by BattingEdge AI Analysis Team.